

Seungju Han

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Machine Learning Engineer with 3+ years of experience building production ML systems across clinical AI, sports analytics, and real-time applications.

PROFESSIONAL EXPERIENCE

ViFiVE, New York, NY

Aug 2022 – May 2025

Machine Learning Engineer

- Led clinical AI validation study with Stanford Medicine, demonstrating vision AI-based ROM measurement accuracy comparable to gold-standard goniometry. Results published in the Journal of Medical Artificial Intelligence (2024).
- Improved **3D pose estimation accuracy by 17%** through a custom self-attention module while maintaining on-device latency requirements for real-time iOS deployment.
- Designed and implemented perspective-agnostic pose system that internally normalizes user orientation, eliminating camera setup requirements and **achieving 70% usability improvement** in beta testing.
- **Enhanced a Random Forest pose classification model accuracy from 89% to 96%** by applying advanced feature engineering techniques, boosting reliability of clinical assessment predictions.
- Optimized data augmentation pipeline and training workflows, **reducing computation overhead by 10%**.

New York Mets, Remote

Oct 2021 – Jul 2022

Data Scientist

- Built **expected distribution model for batted balls with 89% accuracy** by integrating pitcher/batter data including pitch types, handedness, and batted ball statistics.
- Applied K-Means and KNN clustering to identify pitcher archetypes and simulate realistic matchups, **improving player performance forecasting accuracy by 25%**.
- Delivered defensive alignment insights that informed positioning strategy, contributing to the team **improving Defensive Runs Saved from +20 DRS(2021) to +46 DRS(2022)**.

Fastcampus Language, Seoul, Korea

Jun 2020 – Aug 2020

Business Analyst

- Automated KPI dashboard for leadership using Google Suite tools (BigQuery, Looker, and Google Sheets), **reducing reporting time by 75%**.
- Performed SQL-based analysis across multiple teams, identifying trends that contributed to **13% reduction in operating costs**.

PROJECTS

Temporal Clinical Assessment System

- Architected end-to-end ML pipeline for continuous movement analysis, replacing frame-by-frame processing with temporal modeling to capture movement patterns over time.
- Automated functional tests previously requiring manual clinician assessment, improving both convenience and diagnostic accuracy for Seoul National University Hospital.

Human 3D Pose Estimation with Vision Transformers | [Github](#)

- Built 3D human pose estimator using Vision Transformer models, fine-tuned with Low-Rank Adaptation (LoRA).
- Developed a data processing pipeline for handling pose datasets and integrated augmentation strategies.

EDUCATION

University of San Francisco, San Francisco, CA

Jul 2022

MS in Data Science

Relevant Coursework: Machine Learning, Deep Learning, Linear Regression, Time Series Analysis, A/B Testing, Relational Databases, NoSQL, Distributed Computing (Spark), Data Structures & Algorithms

University of California Berkeley, Berkeley, CA

Dec 2019

BA in Statistics

TECHNICAL SKILLS

Languages & ML: Python (NumPy, Pandas), PyTorch, TensorFlow, scikit-learn, Deep Learning, Computer Vision
Data & Infrastructure: SQL (BigQuery, PostgreSQL, MySQL), NoSQL (MongoDB, Firestore), Spark, AWS, GCP, Databricks
Tools: Git, CoreML (iOS deployment), Data Visualization (Matplotlib, Seaborn)